Microcosm: Simulating Life-Like Behaviour in a Real-Time Generative System

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Abstract

Microcosm invites its audience to exist within a fabricated digital habitat, reminiscent of microbiology. Through organic and ever-changing structures, viewers are placed at a molecular point of view, observing life as it might appear beneath an impossible microscope. The work encourages the audience to consider how these same patterns and landscapes exist all around and within us; they simply remain unseen at this scale. The simulation acts as a speculative microscope, submarining into the deepest depths of life to reveal a digital ecosystem that feels simultaneously artificial and alive.

At its core, Microcosm explores how digital systems can mimic organic life through generative design. My central question throughout this project was: **what makes something feel alive?** Rather than representing life directly, I set out to build a system that behaves as though it possesses its own internal logic, something that grows, reacts, and evolves over time.

This project builds on my ongoing exploration of the boundary between digital and organic systems. By blurring those boundaries, Microcosm aims to create an experience that is both futuristic and familiar, a reflection on the hidden ecologies that connect code, matter, and perception.

1. Introduction

This project began with a simple but exacting question: **Can a digital system look and feel biological?** Earlier work of mine relied on static or pre-rendered imagery with occasional 3D experimentations. Here, I wanted liveness, showcased in a generative system that changes in front of the viewer, never quite repeating, that communicates the *tension* between computational order and organic spontaneity.

I was drawn to agent-based particle systems (e.g. Craig Reynold's Boids program to simulate the flocking behavior of birds) and the visual language of micro-landscapes. Those references seeded a conceptual frame: Microcosm would be experienced as if peering through a futuristic microscope, part corporate interface which hints at ethically charged futures where computational and biological life overlap.

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2. Aims and Research Questions

My inspiration came from the intersection of generative art, science fiction, and synthetic biology. I was particularly drawn to artists like **Sage Jenson**, whose work explores emergent complexity through minimal systems. Jenson's '36 Points' piece was especially influential — it demonstrated how simple local rules can lead to global, organic patterns of behaviour.

Beyond that, I looked at works by **Refik Anadol** and **Pierre Huyghe**, both of whom create environments that feel semi-alive, spaces that react, decay, and evolve. These references helped me think about 'Microcosm' not just as a flashy TouchDesigner showcase, but as a world governed by its own laws of physics and perception.

This background gave me the framework to explore questions around authorship and control. How much should I intervene in a system that's meant to appear self-sustaining? When does guidance become interference? These questions guided many of the creative decisions that followed.

A notable important part of researching these artists was looking into how exactly they talk about their work, and providing narrative without having a distinctly beginning-middle-end, hero's journey style of structure. Here, grasping the conceptual narrative provides a seed for an expansive narrative, and can become a feedback loop into the story itself, creating a work that uncovers itself as it is developed and further refined.

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3. Context and Precedents

I explored code examples and tutorials on agent-based systems, learning from both YouTubers and Patreon users. Existing papers by practitioners such as Sage Jenson were useful for discovering how to encode behavioural rules at scale. These references informed my early prototypes, which attempted to build complexity from finding stable parameter values that would create different organic patterns and forms. Depending on the particles behaviour rules, sensor distance, sensor radius, trail length, turn speed and many more, I was able to create a range of different preset values which create differing designs with differentiating logic.

In parallel, I studied microbial imaging as a visual reference point, this included scientific reports, as well as fictional imaginations of the micro-landscape. This allowed me to build

microbial objects utilising shapes, instancing and GLSL code that share the same characteristics of those I was inspired by. I wanted to study how micro-landscapes exhibit movement, colour and atmosphere. Additionally, looking into environments which share the same liminal feeling, space, and deep oceanic planes were inspiring in curating a tone.

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4. Design Methodology and Technical Approach

The technical and methodological direction of 'Microcosm' evolved as a continuation of research I began in the previous semester. That earlier work focused on learning how to build generative systems within **TouchDesigner**, specifically exploring how simple behaviours could lead to complex ecological formations. During that phase, I studied **Sage Jenson's "36 Points"**, learning to implement GLSL shader code inside TouchDesigner. The outcome of that research was a three-minute abstract film that demonstrated how individual particles, when governed by shared rules, could self-organize into living, dynamic structures.

Coming into this semester, my goal was to translate those same emergent behaviours into a **three-dimensional field**. However, I quickly ran into limitations with TouchDesigner's real-time rendering and particle capacity. The scale of 3D space demanded far more agents than the system could efficiently handle, and as a result, the dense organic formations I had achieved in 2D began to thin out. This challenge forced a shift in approach: instead of abandoning the technique, I used it strategically. In the final composition, one scene operates purely on the 3D agent system, while in others, those agents became **background elements** or **positioning data** for instanced geometry.

The organisms that populate *Microcosm* were created through TouchDesigner's instancing workflow, combined with various **form manipulation** and **noise deformation** methods. Each structure began as a simple geometric shape, which I then distorted through procedural modifiers to evoke cellular or microbial behaviour. By layering multiple forms on each other, instancing smaller shapes across larger structures, I was able to achieve more complex creatures. Additionally, applying noise and other textural features, these smooth artificial organisms now were given a roughness that mimicked the surfaces found at a molecular scale.

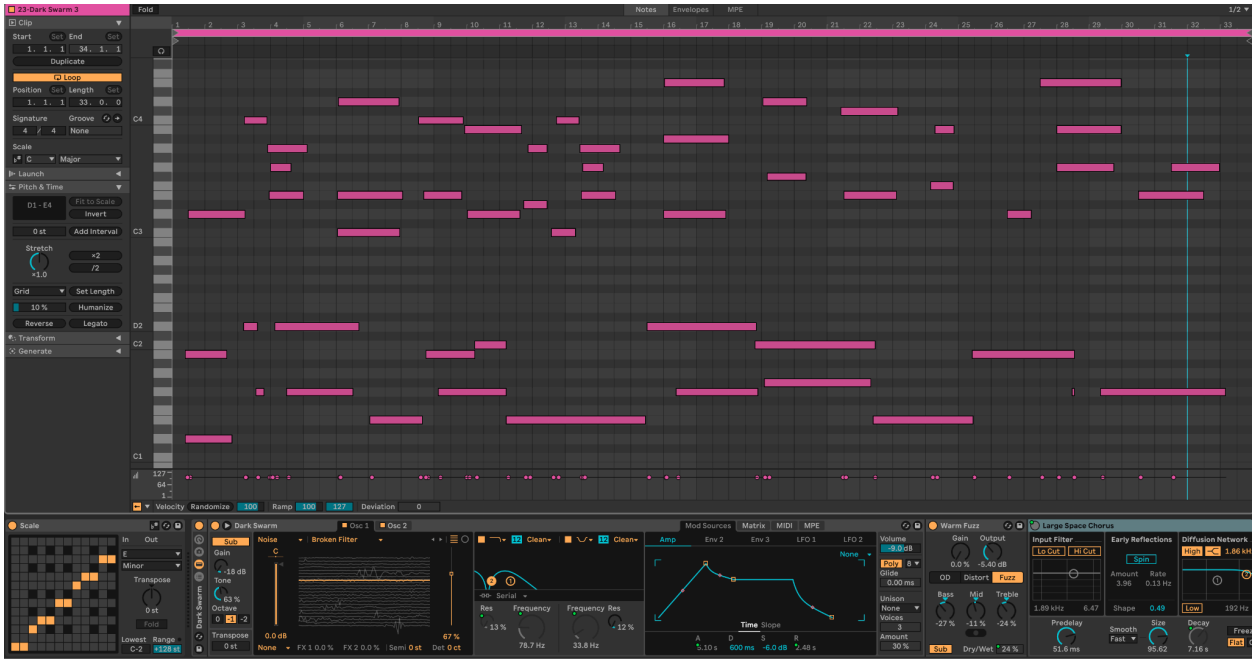
This process naturally shaped the project's tone. The artificiality of the medium became part of the concept: scenes that leaned more electronic and structured reflected the mechanical side of the ecosystem, while others mimicked biological motion and rhythm. That tension between **machine logic and organic flow** became essential in developing the project's identity. It also connected directly to the science-fiction influences underpinning the narrative. *Microcosm* isn't just a real time demonstration, it's a **speculative microscope**, showing a world that feels alive yet unmistakably digital, a glimpse into a possible future where technology and biology have become indistinguishable.

5. Sound Design: Constructing an Auditory Ecosystem

The sound design in *Microcosm* serves not merely as accompaniment but as a **living extension** of the generative system. From the outset, my goal was to create audio that shared the same principles as the visuals in its evolution, unpredictability, and internal logic rather than functioning as a static soundtrack. The process unfolded in **Ableton Live**, where I built layered, evolving soundscapes that mirrored the microbial environments. Sending data back and forth from TouchDesigner, allowed for a design that was reactive, and gave diegetic weight to the objects seen on screen.

5.1 Conceptual Approach

Conceptualising how a microbial landscape sounds was an exciting task, and led to large leaps in the tone and narrative of the piece. This conceptual frame led to sound that felt dense yet spacious, the same way being deep under water carries an overarching feeling of pressure. Oscillating between claustrophobic low-end rumble and delicate, high-frequency textures that resembled biological feedback or electrical interference. Each sound was designed to imply movement or metabolism within the simulated world.



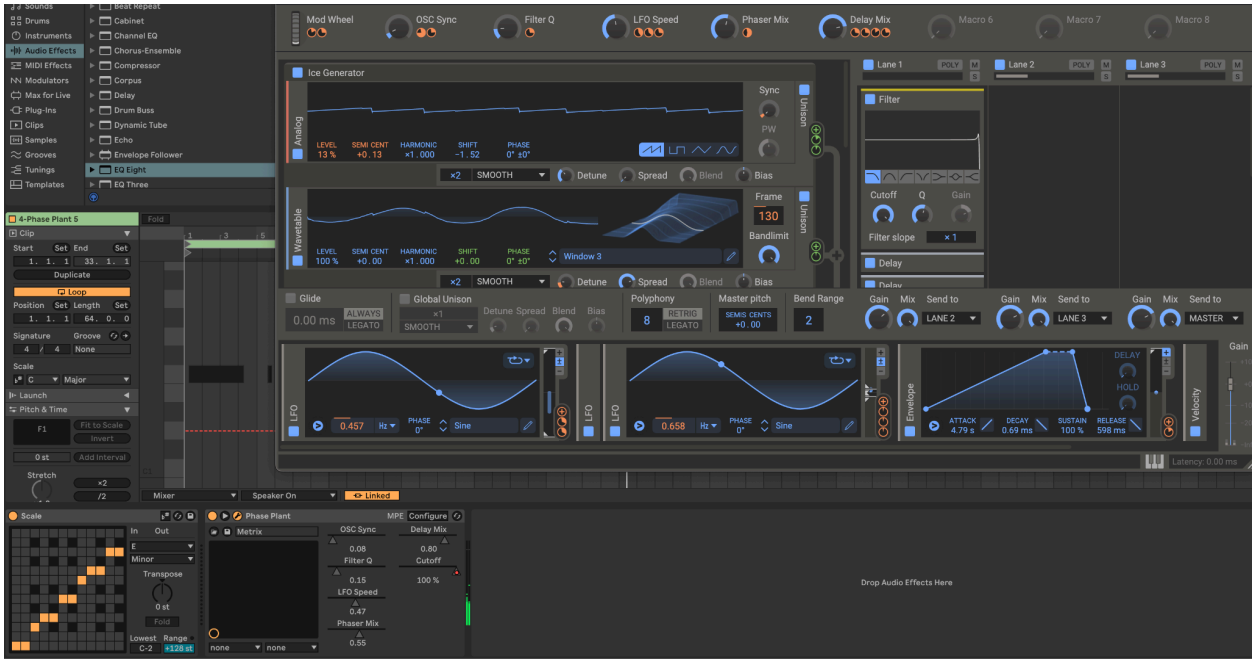
Example of using Ableton's 'Scale' as a way to easily write coherent melodies. By playing off of my keyboard and snapping each key to a specific key. Additionally, utilising Wavetable allows for intricate texture to be applied directly into the synth

5.2 Process and Techniques

The soundscape was constructed from wave-table, granular, and modular synthesis. Using long automation envelopes, LFOs, and randomised modulation, I generated slow-evolving timbres that never perfectly repeated. Noise oscillators were sculpted to create texture and emulate air pressure and micro-movements. Low-end frequencies were key in evoking the sensation of a vast, pressurised environment.

Use of Audio Sampling (recorded an external synth, added modulation and effects)

To maintain a sense of life and autonomy, I used a non-linear structure: no start, build, or drop, just constant evolution. Each element within the session operated on its own cycle length, ensuring that alignments between sounds occurred organically. This made the composition feel like an ongoing process rather than a looped sequence.



Utilising the plugin 'PhasePlant' for its modular synthesis abilities

5.3 Integration with Visuals

The most critical technical aspect was the audiovisual linkage. Audio analysis data was transmitted to TouchDesigner, where it drove subtle visual behaviours as well as making sure the right sound was being played respective to the right scene. By having a soundscape that truly represents and emphasises the visuals, the work suddenly gained tonal and physical weight, creating scenes that felt complete and convincing, without being too obvious to draw an audience away from what's being displayed on scene.

5.4 Iteration and Refinement

Throughout the process, I ran listening tests by letting the environment evolve for extended durations, noting moments where it became predictable or overly dense. I adjusted the behavioural aspects of these sounds, to ensure if given the opportunity, the work could be displayed for an undefined amount of time. I ensured a **clean and balanced mix** by careful EQing, gain staging, and compression, giving each frequency space to breathe while maintaining overall cohesion.

For mastering, I tested Ableton’s built-in mastering presets to determine which enhanced the atmosphere best, then made minor manual adjustments for clarity and headroom. After initial mixing on studio monitors, I refined the final mix in the presentation space itself to account for the room’s acoustic properties, ensuring that the immersive, low-end-driven ambience translated effectively in the black box.

Master Chain